## Client

Static web UI

<<component>>
**React UI**
React + TypeScript
Vite static build

HTTP / JSON

## Spring Boot application — Plane to Notion Connector

One local service process: REST API + scheduled

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**REST controllers**
Configs, projects, jobs

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**Config + catalog**
Provision Notion destinations
Refresh project hierarchy

<<component>>
**Schedule service**
Calculate next occurrence

<<component>>
**Sync job service**
Validate and enqueue requests

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**Scheduled worker**
Poll every 5 seconds
Claim, execute, finalize

<<component>>
**Work-item sync**
Eligibility, mapping, hash
Create or update Notion

<<component>>
**Plane client**
Cursor-paginated reads

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**Notion client**
Pages, data sources, summaries

<<component>>
**Mongo repositories**
Used by app services for persisted state

## External systems

Authenticated REST integrations

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**Plane API**
Workspaces, projects,
work items

<<component>>
**Notion API**
Pages, databases,
data sources

REST

REST

**Notion destination hierarchy**

**Parent page**

-> Workspace page

-> Project page + work-item data source

Workspace and parent summaries roll up project data.

## MongoDB: plane_to_notion

Durable configuration, job state, project catalog, and source-to-target mappings

**sync_configs**
One project per config | options, schedule, status, Notion IDs

**sync_jobs**
Durable queue + run history | trigger, status, timestamps, counts

**sync_mappings**
Plane source ID -> Notion page ID | SHA-256 source hash

**plane_projects + plane_workspaces**
Cached Plane catalog | workspace-to-Notion page link

**notion_destinations**
Created on first task URL sync without a project config

**Index strategy**
Unique project/config and mapping keys; compound job query indexes

## KEY ENGINEERING DECISIONS

**Durable, atomic job processing**
Manual and scheduled work share Mongo job records; atomic claim moves one due job to RUNNING.

**Checkpointed incremental sync**
updated_at bounds plus a 100 ms overlap protect timestamp boundaries between runs.

**Idempotent Notion writes**
Unique mappings and content hashes prevent duplicate pages and skip unchanged writes.

**Bounded data operations**
Plane cursor pages are 200; Mongo mapping lookups and writes are grouped up to 1,000.

**Project-aware destinations**
Each project syncs to its workspace/project Notion hierarchy and contributes to rollups.

**Indexes support hot paths**
Indexes target config uniqueness, job claim/history/checkpoint queries, and mapping uniqueness.